

Accessibility applications

Futuristic embedded systems facilitate numerous accessibility applications for navigating terrains that are rough and difficult to access. Scientists are working on swarm and insect size bots that can navigate vast areas and collect information quickly. At MIT media lab, researchers are working on a flying robot that is trained to track undersea objects such as whales and other marine organisms. Such robots can also be used to clean up toxic waste in chemical industries where it is harmful for human labor. Boston Dynamics created a robot called RHex, a six-legged robot with high mobility and ability to climb rock fields, sand, stairs, water, grass, and other difficult terrains etc. It has IR cameras that enable the operator to control it from far distances.

Smart retail

Today smart phones allow consumers to be connected to the brand and communicate with it in variety of ways. Today it has become imperative for consumer brands to understand the importance of Internet of Things concept and context awareness and apply them in business. With social media applications such as twitter and facebook, consumers stay connected to Brands all the time. Huggies launched an app called TweetPee that helps parents keep track of stock of diapers for their child and order online when needed. Using location meta -data, many brands are working on dynamic pricing of their products. With context aware systems, Brands can create competitive advantages that no other system can give them. There are certain billboards in Japan which instantly alter their displayed messages based on the profile of the consumer set looking at the billboard. This is made possible through image and video processing. Companies such as Tesco in South Korea are working on systems that will define the future of retail shopping. They created virtual stores with products integrated with QR(Quick Response) codes in subways where the busy Korean consumers could just use their smart phones to scan the product QR code and the product will be delivered to their doorstep.

Security and defense

Since long, one major application and research area for embedded systems has been security and defense, and even in future some of the most promising developments will be seen in this area. On simple application, “Capture Resistant Environment” is being developed at UbiComp lab in University of Washington.